

Exponential dispersion models

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Last time: Exponential dispersion models

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Example $Y \sim N(\mu, \sigma^2)$ $f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y-\mu)^2 \right\}$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} y^2 \right\} \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} \right\}$$

$$\theta = \mu$$

$$\kappa(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$$

$$\phi = \sigma^2$$

Exponential dispersion models

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Example $Y \sim \text{Bernoulli}(p)$

$$f(y; p) = p^y (1-p)^{1-y} = \exp \left\{ \log(p^y (1-p)^{1-y}) \right\}$$

$$= \exp \left\{ y \log(p) + (1-y) \log(1-p) \right\}$$

$$= \exp \left\{ y \log \left(\frac{p}{1-p} \right) + \log(1-p) \right\}$$

$$\theta = \log \left(\frac{p}{1-p} \right)$$

$$\kappa(\theta) = -\log(1-p) = -\log \left(1 - \frac{e^\theta}{1+e^\theta} \right) = \log(1+e^\theta)$$

$$\phi = 1$$

$$a(y, \phi) = 1$$

EDM components

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

θ : canonical parameter

$$\theta = g(\mu) \quad \mu = E[Y]$$

$\uparrow$ canonical link function

ϕ : dispersion parameter (> 0)

$$\text{Var}(Y) = \phi \cdot \underbrace{v(\mu)}_{\text{describing shape of mean-variance relationship}}$$

Ex: $N(\mu, \sigma^2)$: $\text{Var}(Y) = \sigma^2 = \sigma^2 \cdot 1$
 $\phi = \sigma^2 \quad v(\mu) = 1$

Poisson(μ) : $\text{Var}(Y) = \mu = 1 \cdot \mu$
 $\phi = 1 \quad v(\mu) = \mu$

Bernoulli(μ) : $\text{Var}(Y) = \mu(1-\mu) = 1 \cdot \mu(1-\mu)$
 $\phi = 1 \quad v(\mu) = \mu(1-\mu)$

EDM components

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

$\kappa(\theta)$: cumulant function

$$\boxed{\frac{d\kappa(\theta)}{d\theta} = \mu}$$

Ex: $N(\mu, \sigma^2)$: $\theta = \mu$ $\kappa(\theta) = \frac{\theta^2}{2}$
 $\frac{d}{d\theta} \kappa(\theta) = \theta = \mu$

Poisson(μ): $\theta = \log(\mu)$ $\kappa(\theta) = \mu = e^\theta$
 $\frac{d}{d\theta} \kappa(\theta) = e^\theta = \mu$

$$\boxed{\frac{d^2}{d\theta^2} \kappa(\theta) = v(\mu)}$$

Ex: $N(\mu, \sigma^2)$ $\kappa(\theta) = \frac{\theta^2}{2}$ $\frac{d^2}{d\theta^2} \kappa(\theta) = 1$

Poisson(μ) $\kappa(\theta) = e^\theta$ $\frac{d^2}{d\theta^2} \kappa(\theta) = e^\theta = \mu$

Cumulants and the cumulant generating function

Cumulant generating function (CGF) :

MGF : $M(t) = \mathbb{E}[e^{tY}] \quad \frac{d^r}{dt^r} M(t) \Big|_{t=0} = \mathbb{E}[Y^r]$

CGF : $C(t) = \log M(t) = \log \mathbb{E}[e^{tY}]$

activity $\begin{cases} \frac{d}{dt} C(t) \Big|_{t=0} = \mathbb{E}[Y] \\ \frac{d^2}{dt^2} C(t) \Big|_{t=0} = \text{Var}(Y) \end{cases}$

For an EDM:

$$C(t) = \frac{\kappa(0 + t\theta) - \kappa(0)}{\theta}$$

Cumulants and the cumulant generating function

For an EDM:

$$C(t) = \frac{\kappa(\theta + t\phi) - \kappa(\theta)}{\phi}$$

$$\begin{aligned}\text{So: } \frac{d}{dt} C(t) \Big|_{t=0} &= \frac{\kappa'(\theta + t\phi)}{\phi} \cdot \phi \Big|_{t=0} \\ &= \kappa'(\theta + t\phi) \Big|_{t=0} = \kappa'(\theta)\end{aligned}$$

$$\therefore E[Y] = \kappa'(\theta)$$

$$\begin{aligned}\frac{d^2}{dt^2} C(t) \Big|_{t=0} &= \frac{d}{dt} \kappa'(\theta + t\phi) \Big|_{t=0} \\ &= \kappa''(\theta + t\phi) \phi \Big|_{t=0} \\ &= \phi \kappa''(\theta)\end{aligned}$$

$$\begin{aligned}\therefore \text{Var}(Y) &= \phi \kappa''(\theta) = \phi \frac{d\mu}{d\theta} \\ &= \phi V(\mu)\end{aligned}$$

Since $\text{Var}(Y) > 0$ & $\phi > 0$, $\Rightarrow \frac{d\mu}{d\theta} > 0 \Rightarrow \mu \text{ \& } \theta \text{ are 1-to-1, increasing functions of each other}$

Cumulants and the cumulant generating function

For an EDM:

$$C(t) = \frac{\kappa(\theta + t\theta) - \kappa(\theta)}{\theta}$$

$$\begin{aligned} m(t) &= \mathbb{E}[e^{ty}] = \int_{\mathcal{Y}} e^{ty} a(y, \theta) \exp\left\{\frac{y\theta - \kappa(\theta)}{\theta}\right\} dy \\ &= \int_{\mathcal{Y}} a(y, \theta) \exp\left\{\frac{y\theta - \kappa(\theta) + ty\theta}{\theta}\right\} dy \end{aligned}$$

$$\begin{aligned} \text{Let } \theta^* &= \theta + t\theta \\ \Rightarrow m(t) &= \int_{\mathcal{Y}} a(y, \theta) \exp\left\{\frac{y\theta^* - \kappa(\theta)}{\theta}\right\} dy \\ &= \int_{\mathcal{Y}} a(y, \theta) \exp\left\{\frac{y\theta^* - \kappa(\theta^*)}{\theta}\right\} \exp\left\{\frac{\kappa(\theta^*) - \kappa(\theta)}{\theta}\right\} dy \\ &= \exp\left\{\frac{\kappa(\theta^*) - \kappa(\theta)}{\theta}\right\} \int_{\mathcal{Y}} a(y, \theta) \exp\left\{\frac{y\theta^* - \kappa(\theta^*)}{\theta}\right\} dy \\ &= \exp\left\{\frac{\kappa(\theta^*) - \kappa(\theta)}{\theta}\right\} = \exp\left\{\frac{\kappa(\theta + t\theta) - \kappa(\theta)}{\theta}\right\} \\ \therefore C(t) &= \log m(t) = \frac{\kappa(\theta + t\theta) - \kappa(\theta)}{\theta} \end{aligned}$$

Activity

Work on the activity (handout).